STA 141 Project

Soccer Injury Analysis

by

Roman Vasilyev

Brief Overview

Injuries are a central concern in professional soccer, as they directly impact player availability, team performance, and long-term career sustainability. Because different positions on the field involve unique physical demands, players face different injury risks. For instance, defenders are often exposed to high-contact collisions, while forwards are more likely to experience sprint-related muscle injuries. The goal of this project is to apply data analysis methods to identify which positions are most prone to injuries, determine the most common injury types by position, and interpret why these patterns occur. Understanding these trends has practical implications for player health management, coaching strategies, and resource allocation in sports medicine.

The dataset used in this project is a publicly available soccer injury database in CSV format, sourced from Kaggle. The file, *player_injuries_impact.csv*, contains detailed records of injuries across multiple seasons and players. Key variables include player position, age, and injury type (e.g., hamstring, knee, ankle). This rich dataset makes it possible to study position-specific injury patterns in depth.

Key Questions:

1. Which positions in soccer experience the highest frequency of injuries?
2. What are the most common types of injuries overall, and how do they vary across positions?
3. Are injury types significantly different across age?

Data Manipulation

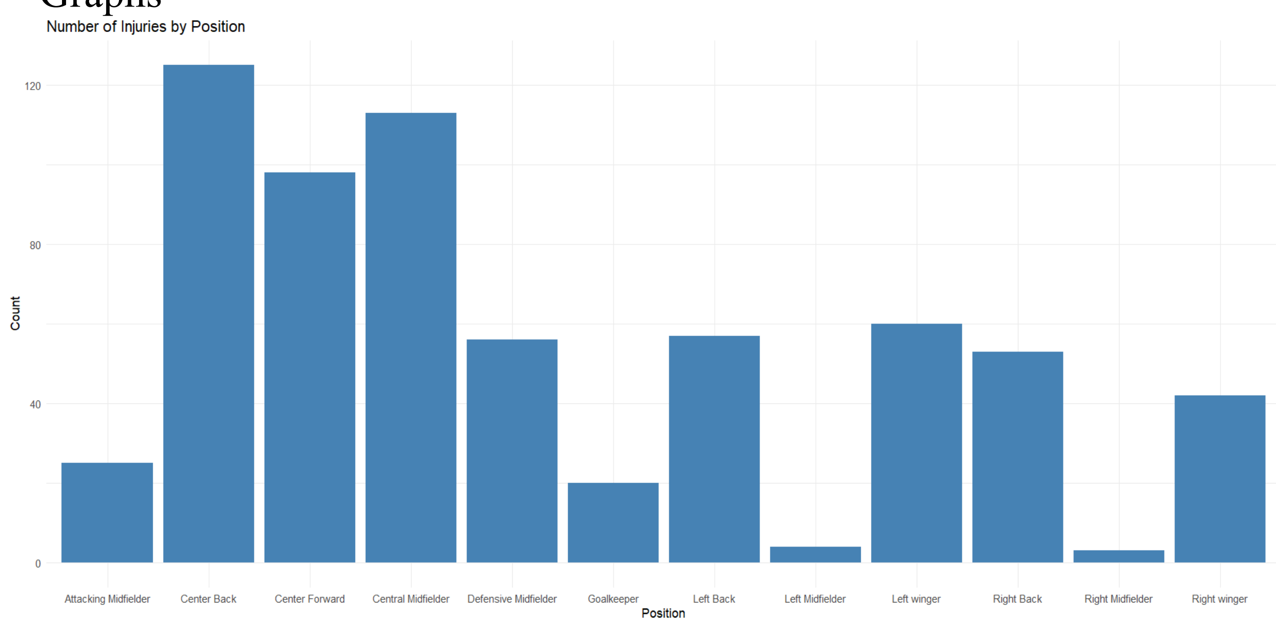
After selecting the correct dataset, importing it into R, and tidying it up, I created a custom injury rate function to calculate the total number of injuries per position along with the average age of players in each group.

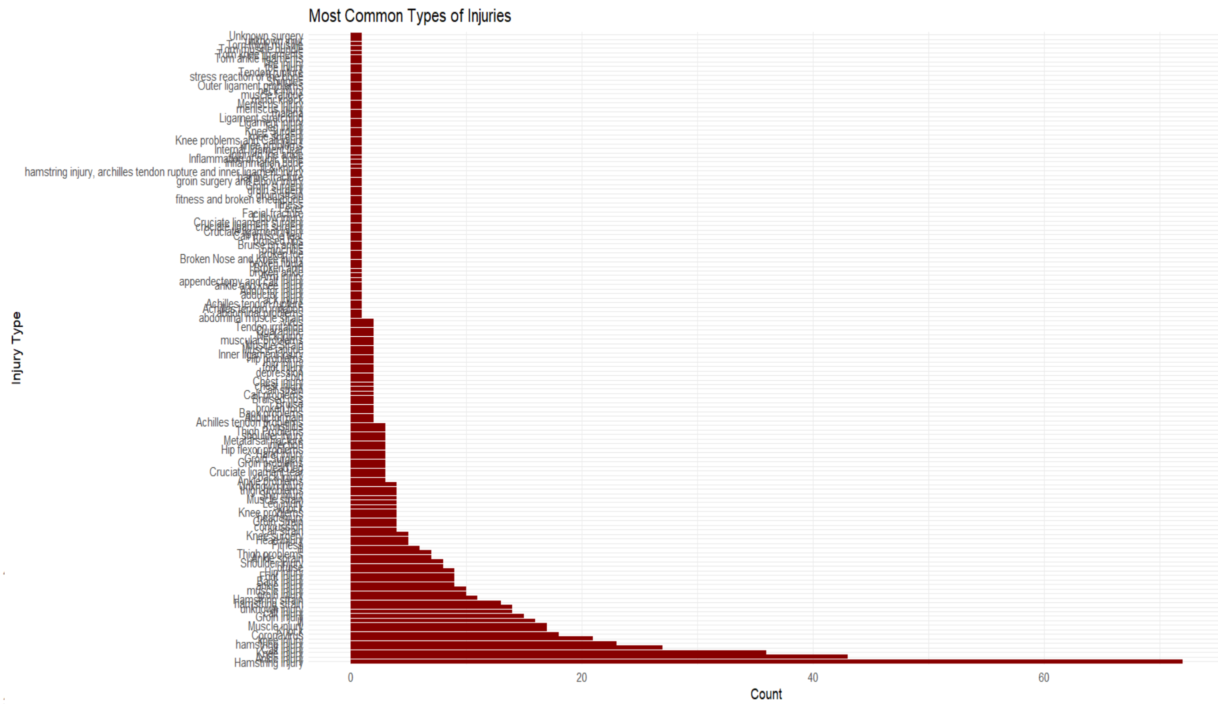
```
position              total_injuries avg_age
   <chr>                        <int>   <dbl>
 1 Center Back                    125    27.2
 2 Central Midfielder             113    26.1
 3 Center Forward                  98    28.2
 4 Left winger                     60    24.3
 5 Left Back                       57    25.5
```

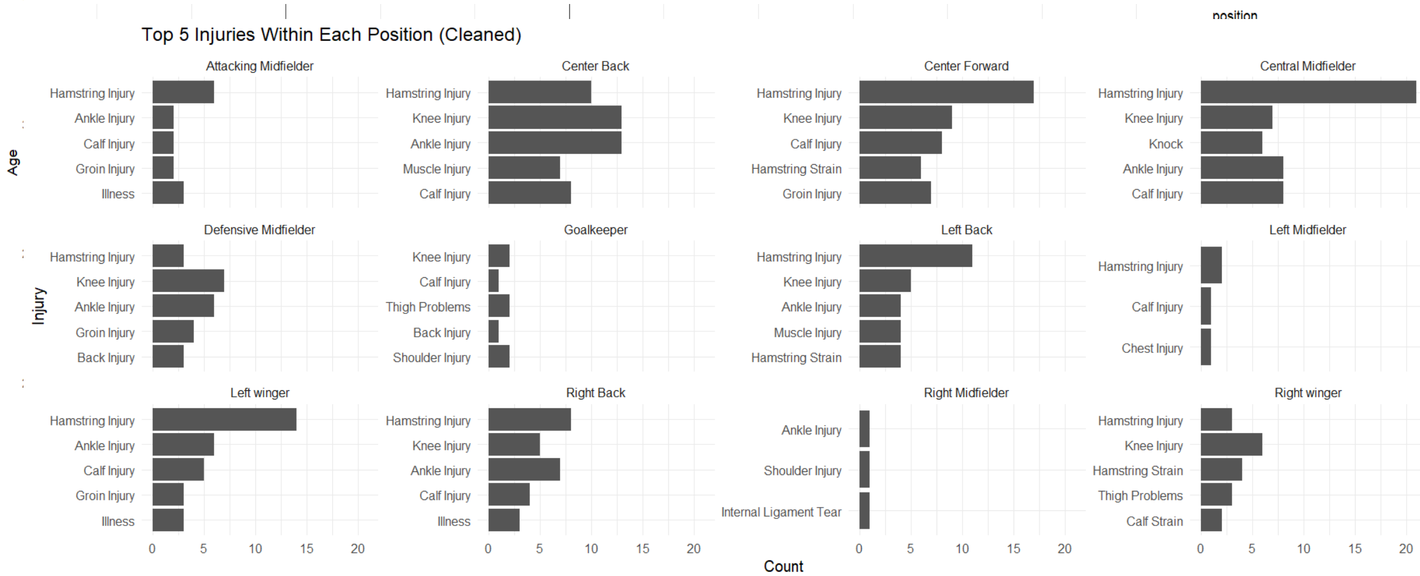
6	Defensive Midfielder	56	26.8
7	Right Back	53	26.2
8	Right winger	42	26.5
9	Attacking Midfielder	25	26.4
10	Goalkeeper	20	30.0
11	Left Midfielder	4	28.2
12	Right Midfielder	3	30.3

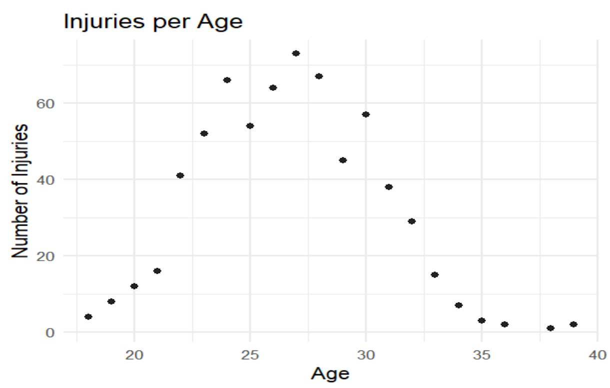
The results show that center backs, central midfielders, and center forwards suffer the highest number of injuries, suggesting that players in central positions are most prone to injury overall. This makes intuitive sense. As a soccer player myself, I've observed that play in the middle of the field tends to be more contested, with more tackles and physical contact, while wide players are often less involved in heavy collisions.One interesting observation is that left and right midfielders appear as outliers, with very low reported injuries compared to expectations. In reality, wide midfielders usually experience injury frequencies closer to defensive midfielders. This discrepancy may suggest underreporting or gaps in the dataset for these positions. Another fact to note is that right midfielders and right wingers may be considered the same. Depending on the formation that the team is playing, there may be right midfielders. I am thinking that the data set contains teams that play more wingers than midfielders.Looking at the average age, there are some clear patterns. Wide attacking players such as left wingers are the youngest group to experience injuries (average age 24.3), which may reflect the speed and physical demands of wing play on younger players. By contrast, goalkeepers and right midfielders have the highest average ages at 30.0 and 30.3 years. Older players may be more susceptible to wear-and-tear injuries or take longer to recover, which could explain their higher average ages despite relatively low injury counts. Central defenders and forwards, who top the injury list, average around 27–28 years old, right in the prime of a soccer career suggesting that injury risk peaks during peak playing years rather than only at very young or very old ages.

Graphs




Most Common Types of Injuries
Injury Type
Count
0
20
40
60
Unknown surgery
Torn ankle ligaments
stress reaction of the bone
Outer ligament problems
Ligament stretching
Knee problems
hamstring injury, archilles tendon rupture and inner ligament injury
fitness and broken cheekbone
Facial fracture
Cruciate ligament surgery
Broken Nose and Knee Injury
appendectomy
abdominal muscle strain
Inner ligament
Achilles tendon problems
Hip flexor problems
Cruciate ligament tear
Knee problems
Thigh problems
Hamstring Strain
Muscle injury
Hamstring Injury


Top 5 Injuries Within Each Position (Cleaned)
Injury
Count
Attacking Midfielder
Hamstring Injury
Ankle Injury
Calf Injury
Groin Injury
Illness
Center Back
Hamstring Injury
Knee Injury
Ankle Injury
Muscle Injury
Calf Injury
Center Forward
Hamstring Injury
Knee Injury
Calf Injury
Hamstring Strain
Groin Injury
Central Midfielder
Hamstring Injury
Knee Injury
Knock
Ankle Injury
Calf Injury
Defensive Midfielder
Hamstring Injury
Knee Injury
Ankle Injury
Groin Injury
Back Injury
Goalkeeper
Knee Injury
Calf Injury
Thigh Problems
Back Injury
Shoulder Injury
Left Back
Hamstring Injury
Knee Injury
Ankle Injury
Muscle Injury
Hamstring Strain
Left Midfielder
Hamstring Injury
Calf Injury
Chest Injury
Left winger
Hamstring Injury
Ankle Injury
Calf Injury
Groin Injury
Illness
Right Back
Hamstring Injury
Knee Injury
Ankle Injury
Calf Injury
Illness
Right Midfielder
Ankle Injury
Shoulder Injury
Internal Ligament Tear
Right winger
Hamstring Injury
Knee Injury
Hamstring Strain
Thigh Problems
Calf Strain
0
5
10
15
20



The bar chart of injuries by position shows that center backs, central midfielders, and center forwards experience the highest number of injuries, while wide players such as fullbacks, wingers, and midfielders have fewer overall. This reinforces the idea that the middle of the pitch is more contested and physically demanding, putting central players at greater risk. The second chart, displaying injury types, suggests that certain injuries dominate the dataset, particularly hamstring strains, knee problems, and cruciate ligament injuries. Finally, the boxplot of age distribution shows that most positions cluster around the mid-20s, with interquartile ranges from roughly 23 to 29 years old. Goalkeepers, central forwards, and wide midfielders tend to be older on average, while wingers skew younger, often experiencing injuries in their early 20s. I would retest this with another data set because intuitively I would assume that with age there are higher amounts of injuries. Outliers appear across most positions, representing players either much younger or older than the typical age range. The breakdown of injuries by position shows that hamstring injuries dominate across nearly all roles, especially for wingers and central midfielders, reflecting the sprinting and high-intensity demands of those positions. Knee and ankle injuries are also common, particularly in central and defensive roles where tackling and pivoting are frequent. Goalkeepers stand out with fewer overall injuries but a different profile dominated by back and shoulder problems, consistent with the physical demands of diving and repetitive strain. Left and right midfielders appear as outliers with very few recorded injuries, likely due to underrepresentation in the dataset. Overall, central players and wingers face the highest risk of lower-body injuries, while centerbacks and defenders face the highest risk of knee and ankle injuries.

Data Analysis

`Q1. Which positions experience the highest frequency of injuries?`

```
> lm_q1 <- lm(injury_count ~ avg_age, data = injury_by_pos)
> summary(lm_q1)

Call:
lm(formula = injury_count ~ avg_age, data = injury_by_pos)

Residuals:
   Min     1Q Median     3Q    Max 
-40.54 -21.05 -11.35  10.77  71.10 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  302.067    178.893   1.689    0.122
avg_age       -9.116      6.579  -1.386    0.196

Residual standard error: 38.58 on 10 degrees of freedom
Multiple R-squared:  0.1611,	Adjusted R-squared:  0.07718 
F-statistic:  1.92 on 1 and 10 DF,  p-value: 0.196

> tidy(lm_q1, conf.int = TRUE)
# A tibble: 2 × 7
  term        estimate std.error statistic p.value conf.low conf.high
  <chr>          <dbl>     <dbl>     <dbl>   <dbl>    <dbl>     <dbl>
1 (Intercept)   302.       179.       1.69   0.122    -96.5    701.  
2 avg_age        -9.12       6.58    -1.39   0.196    -23.8      5.54
> glance(lm_q1)
# A tibble: 1 × 12
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC   BIC deviance df.residual  nobs
      <dbl>         <dbl> <dbl>     <dbl>   <dbl> <dbl>  <dbl> <dbl> <dbl>    <dbl>       <int> <int>
1     0.161        0.0772  38.6      1.92   0.196     1  -59.8  126.  127.   14886.          10    12
```

Q2. What are the most common injuries overall, and how do they vary across positions?

Models: Linear Probability Models (LPM) for each of the Top-5 injuries:

```
> lpm_pos_coef  # position effects for each top injury (coefficients are changes in probability)
# A tibble: 60 × 8
   term                        estimate std.error statistic p.value conf.low conf.high injury_target 
   <chr>                          <dbl>     <dbl>     <dbl>   <dbl>    <dbl>     <dbl> <chr>         
 1 (Intercept)                  0.0800     0.0311     2.57  0.0104   0.0189     0.141  Hamstring Injury
 2 positionCentral Midfielder   0.106      0.0452     2.34  0.0194   0.0172     0.195  Hamstring Injury
 3 positionCenter Forward       0.0935     0.0469     1.99  0.0469   0.00130    0.186  Hamstring Injury
 4 positionLeft winger          0.153      0.0546     2.81  0.00516  0.0460     0.261  Hamstring Injury
 5 positionLeft Back            0.113      0.0556     2.03  0.0426   0.00380    0.222  Hamstring Injury
 6 positionDefensive Midfielder -0.0264    0.0559    -0.472 0.637   -0.136      0.0834 Hamstring Injury
 7 positionRight Back           0.0709     0.0570     1.24  0.214   -0.0410     0.183  Hamstring Injury
 8 positionRight winger        -0.00857    0.0620    -0.138 0.890   -0.130      0.113  Hamstring Injury
 9 positionAttacking Midfielder 0.160      0.0762     2.10  0.0362   0.0103     0.310  Hamstring Injury
10 positionGoalkeeper          -0.0800     0.0838    -0.955 0.340   -0.245      0.0845 Hamstring Injury
# ℹ 50 more rows
# ℹ Use `print(n = ...)` to see more rows
> lpm_pos_fit   # model fit stats per injury (R^2, F-test, etc.)
# A tibble: 5 × 13
  r.squared adj.r.squared sigma statistic p.value    df logLik    AIC    BIC deviance df.residual  nobs
      <dbl>         <dbl> <dbl>     <dbl>   <dbl> <dbl>  <dbl>  <dbl>  <dbl>    <dbl>       <int> <int>
1   0.0407        0.0243  0.348     2.48  0.00469    11 -232.    490.   549.     77.9         644   656
2   0.0125       -0.00434 0.283     0.743 0.698      11  -95.5   217.   275.     51.4         644   656
3   0.0193        0.00257 0.270     1.15  0.317      11  -65.9   158.   216.     47.0         644   656
4   0.0120       -0.00490 0.243     0.710 0.730      11    3.71   18.6   76.9    38.0         644   656
5   0.00743      -0.00953 0.200     0.438 0.939      11  132.   -238.  -179.     25.7         644   656
# ℹ 1 more variable: injury_target <chr>
```

```
Call:
glm(formula = y_ankle ~ is_def, family = binomial(), data = df)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)    
(Intercept)  -2.4321     0.1789 -13.597   <2e-16 ***
is_def        0.4715     0.2671   1.765   0.0775 .  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 414.97  on 655  degrees of freedom
Residual deviance: 411.90  on 654  degrees of freedom
AIC: 415.9

Number of Fisher Scoring iterations: 5

> summary(fit_knee)

Call:
glm(formula = y_knee ~ is_def, family = binomial(), data = df)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)    
(Intercept)  -2.2266     0.1644 -13.547   <2e-16 ***
is_def        0.2660     0.2576   1.033    0.302    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 445.52  on 655  degrees of freedom
Residual deviance: 444.47  on 654  degrees of freedom
AIC: 448.47

Number of Fisher Scoring iterations: 4
```

```
# Q3. Does age affect the type of injury? Models: Linear Probability Models
(LPM) for each of the Top-5 injuries:
```

```
# A tibble: 10 × 8
   term        estimate std.error statistic  p.value  conf.low conf.high injury_target
   <chr>          <dbl>     <dbl>     <dbl>    <dbl>     <dbl>     <dbl> <chr>
 1 (Intercept)  0.217     0.103       2.10  0.0360    0.0143    0.421    Hamstring Injury
 2 age         -0.00272   0.00385    -0.708 0.479    -0.0103    0.00483  Hamstring Injury
 3 (Intercept)  0.254     0.0826      3.08  0.00216   0.0922    0.416    Knee Injury
 4 age         -0.00628   0.00307    -2.05  0.0412   -0.0123   -0.000252 Knee Injury
 5 (Intercept)  0.290     0.0790      3.67  0.000260  0.135     0.445    Ankle Injury
 6 age         -0.00791   0.00294    -2.69  0.00726  -0.0137   -0.00214  Ankle Injury
 7 (Intercept) -0.102     0.0709     -1.44  0.150    -0.241     0.0369   Calf Injury
 8 age          0.00618   0.00263     2.35  0.0193    0.00101   0.0114   Calf Injury
 9 (Intercept) -0.0609    0.0583     -1.05  0.296    -0.175     0.0535   Muscle Injury
10 age          0.00383   0.00217     1.77  0.0777   -0.000426  0.00808  Muscle Injury
> lpm_age_fit   # model fit stats per injury
# A tibble: 5 × 13
  r.squared adj.r.squared sigma statistic p.value    df logLik     AIC    BIC deviance df.residual  nobs
      <dbl>         <dbl> <dbl>     <dbl>   <dbl> <dbl>  <dbl>   <dbl>  <dbl>    <dbl>       <int> <int>
1  0.000766     -0.000762 0.352     0.501 0.479       1 -245.    497.   510.       81.2         654   656
2  0.00636       0.00484  0.281     4.19  0.0412      1  -97.6   201.   215.       51.7         654   656
3  0.0110        0.00946  0.269     7.25  0.00726     1  -68.7   143.   157.       47.4         654   656
4  0.00834       0.00682  0.241     5.50  0.0193      1    2.50    0.998  14.5     38.1         654   656
5  0.00475       0.00323  0.198     3.12  0.0777      1  131.   -256.   -242.      25.8         654   656
# ℹ 1 more variable: injury_target <chr>
```

I limited the data to age, position, and injury, standardized injury labels with regular expressions (so “hamstring injury” variants line up), and created simple helper features

Model 1 — OLS on position-level counts. (injury_count ~ avg_age)

Why: checks whether positions that skew older also rack up more injuries.

Model 2 — Linear Probability Models (position → injury type).
Why: tests whether certain positions are more likely to suffer a given injury type.

Model 3 — Linear Probability Models (age → injury type).
Why: checks simple age trends for each injury category.

Model 4 — Logistic regression (defenders vs. others).
Why: isolates the tackling/change-of-direction hypothesis for back-line players.

Across 656 injury records and 12 positions, the most injury-prone roles are Center Back (125 cases), Central Midfielder (113), and Center Forward (98), with the remaining positions ranging from 60 down to 3 cases. A simple OLS of position-level counts on average age (injury_count ~ avg_age) yields a negative but non-significant slope (−9.12 injuries per additional year; $p = 0.196$, $R^2 = 0.16$), indicating no reliable association between a position's average age and how often injuries occur. For injury types, the top five are Hamstring, Knee, Ankle, Calf, and Muscle. Linear probability models of 1[injury = type] on position show that position meaningfully explains hamstring injuries (overall F-test $p = 0.0047$, $R^2 \approx 4.1\%$): compared with Center Backs, Central Midfielders (+10.6 percentage points), Center Forwards (+9.3 pp), Left Wingers (+15.3 pp), Left Backs (+11.3 pp), and Attacking Midfielders (+16.0 pp) have higher hamstring injury probabilities (all $p < 0.05$). For knee, ankle, calf, and generic muscle injuries, position alone does not explain much variation (non-significant overall tests).

Age effects estimated with simple LPMs (1[injury = type] ~ age) suggest a small but systematic pattern. That is the probability of knee (−0.0063 per year, $p = 0.041$) and ankle (−0.0079 per year, $p = 0.0073$) injuries decreases with age, while calf injuries increase (+0.0062 per year, $p = 0.019$); "muscle" is borderline (+0.0038 per year, $p = 0.078$) and hamstring shows no detectable age effect (−0.0027, $p = 0.479$). Effect sizes are modest ($R^2 \leq \sim 1\%$), so these age relationships are statistically detectable but practically small in this sample. To test defenders specifically, single-predictor logistics (no age) show defenders have higher ankle odds and slightly higher knee odds. Age effects are small overall. The LPMs, knee and ankle probabilities decline with age while calf rises, and hamstrings show no age trend. Hamstrings are the headline and are position-sensitive. Defenders trend higher on ankle (and modestly knee) injuries. Age patterns exist but are weak in this sample.

Conclusion

The regressions largely back up our claims about hamstrings, with a nuance. Hamstring injuries are the most frequent in this sample and, unlike other types, they show clear positional differences in midfield and attacking roles where Central Midfielder, Center Forward, Left Winger, Left Back, Attacking Midfielder have significantly higher hamstring risk relative to Center Backs. It makes sense that Defenders are more prone to knee and ankle injuries as those parts of the body are prone to damage when tackling. Thinking about the style of play is important in this case. Attackers are more prone to hamstring injuries as they need to quickly burst off and switch directions. Where our analysis underperforms is on age. Based on personal experience, and following soccer trends, with age there are higher amounts of injuries. Unfortunately, the data set is not diverse enough to represent this.